Vakint reduction benchmark

Cached RustRed versus the FORM / FMFT reduction route

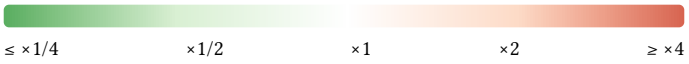
2026-10-24 · AMD EPYC 9754 128-Core Processor · CPU affinity 57

Reading guide. Bold values are RustRed times; FMFT times appear below. Multipliers are *RustRed ÷ FMFT*: green means faster, white parity, red slower. Both receive identical four-loop scalar inputs. The first physical denominator has power 2 (base) or 4 (held out); all others have power 1, auxiliaries 0.

Reduction time

Median seconds over 3 fresh processes per backend and case.

Topology	Base · power 2	Held out · power 4
H	**0.0359 s** FMFT 0.670 s ×0.054	**3.98 s** FMFT 7.37 s ×0.54
FG	**0.0450 s** FMFT 0.217 s ×0.21	**0.112 s** FMFT 0.430 s ×0.26
BMW	**0.0235 s** FMFT 0.181 s ×0.13	**0.997 s** FMFT 3.30 s ×0.30
X	**0.0822 s** FMFT 1.61 s ×0.051	**15.8 s** FMFT 21.0 s ×0.75



Method and correctness. Build profile: release (caller-declared); optimization/LTO settings are not inferred. rustc 1.98.1; FORM 5.0.0. Both retain masters in FMFT/MATAD normalization, with identical symbolic mass. Exact rational parity passed all eight cases at unit mass and renormalization scale, without truncating epsilon or using numerical tolerances.

Base cases hit prepared roots without applying rules. Held-out cases reuse cached descendants after root misses. Timing covers first public reduction calls, including FORM version checks, launch, I/O and parsing. Data loading, cache attachment, master evaluation and validation are excluded. Fresh processes run sequentially, alternating backends, with the recorded CPU affinity on a shared host. Filesystem pages are warm; this does not measure cold startup.

Selected validated run: `2026-10-24T06:59:51Z-8da99b9a5067`. Source: `8ef03d7fd755`. Data: `v0.1.2`. Samples and provenance: `data/results.json`. Refresh instructions: `../../AGENTS.md`.

Five-loop validation and benchmarks

Retained-master comparisons · functional evidence reported separately

Historical DOT ingress acceptance

2026-10-23 · one sequential debug-build pass. This checks graph definitions; it does not time IBP reduction, master evaluation, package loading or cache preparation.

Validation workload	Recorded result
Master graphs / dotted / with numerator	640 / 441 / 204
Exact routing, full numerators, native ingress and DOT round trips	Passed
Complete graph-validation pass	61.8 s
Per graph: minimum / median / maximum	0.0443 / 0.0955 / 0.175 s
Numerical master values / prepared 5L cache in this run	Absent / absent

Physical powers, signed indices, mass terms, full numerators, loop-basis slots (including zero) and integration Jacobians were checked exactly. These checks establish the stored graph definitions, without claiming that the catalogue is an independent master basis. One observed run supports no statistical speedup claim. Earlier interrupted checks are not a timing baseline.

Five-loop reduction comparison

Retained-master workload	No persistent cache	With persistent cache
Complete quadratic UV fixture	Deferred	Deferred
Zero-momentum diagnostic	Pending	Pending
At least 12 held-out inputs	Pending	Pending

Eligibility and reading guide. Primary cells require catalogue closure and exact result/guard agreement, a completed strict cache, and at least three matched fresh processes per mode in one accepted serialized window, with fixed workers and affinity. Bold values are reduction seconds; small values give the full observed range. Multipliers are *cached ÷ uncached*, using the same colour scale as page 1. Functional runs, including the exploratory q0 cohort, never supply these medians. Cache preparation and held-out performance retain separate evidence. The user waived new uncached q0 repetitions; its historical reference and completed functional checks appear on the delivery page.

Historical ingress: source `68f2263bf54d` (modified); RustRed `52c7463686b4` Symbolica `b6cca776f9dc`. Exact patch, executable, data and per-graph digests remain in JSON.

Historical compatibility record, separate from selected performance evidence: RustRed `91b72f8f5d42` Symbolica `9c8623ecb800`. Published upgrade: Symbolica `1205ec30890f`, FeynKit `d22e0f9fdbf8` acceptance was pending in that historical record. Full revisions: `data/results.json`.

Five-loop evidence and timing ledger

Selected report: `five-loop-2f0a5a637cf939141c8b1eeb`. Raw reports and their hashed launcher/compiler receipts remain in JSON. Qualified performance additionally binds complete result/guard files, unknown-master inventories and accepted window receipts. A successful retained expression alone does not establish catalogue closure or numerical evaluability.

Primary workloads

Complete quadratic UV fixture: Deferred; unmeasured in this scope.

Zero-momentum diagnostic: pending.

Finite selected-cache preparation

Producer `8ef03d7fd755` · original phase: `pilot`. Retained completed candidate; selection recorded separately as `828f61d0d6a6`. 2 / 2 selected roots complete; 25086 intermediate records; strict Reject policy. Compressed size: 2.342 MiB; decoded: 16593448 bytes (15.825 MiB). Original selected-pilot producer-driver elapsed: `1272.632297156` s, including loading and owner setup, not pure IBP or overall campaign time. Cache SHA-256: `cc5e6d80aacc737b81be1675a8aa02694b9ccaa1e4f31cc2d5a34199e4f1a19d`. Training roots, capture diagnostics and source/proof bindings remain in the unmodified build receipt.

Accepted native compatibility bridge `daa07861eaf7` · consumer cohort receipt `2b2a26eb74cc` · consumer workload `f022d2173668`. This does not establish a fresh target-suite pass, later-cohort compatibility or performance.

Separate rejected trial: 1 / 3 roots complete; `32613.349550541956` s whole attempt; no final gzip candidate.

This finite selection retains its original producer and acceptance. The cumulative cache from physical-example reductions is a separate record; see the physical capture supplement.

Timing boundaries. Stages overlap; never sum them. Owner preparation is included in saved-application/reduction times. `ibp_excluding_owner_seconds` means saved application minus owner preparation, including transport, combination and terminal substitution. Pure IBP, terminal-only, isolated gzip decode and primary owner-subtracted times remain unmeasured. Prepared-input E2E includes current input/data/cache loading and public reduction; generator preparation stays historical. Fresh/prepared scopes never mix. Initial loading uses Sparse; representation reload belongs to reduction. Missing events are "—", never zero. Full records remain in JSON.

Genuinely connected five-loop control

Supplement: same-engine warm reuse. This scalar ladder-pinch control is separate from the physical quadratic/q0 workloads and from held-out persistent-cache comparisons. Each fresh process makes one first public API call and then repeats it on the same engine, settings and input.

1 completed functional pair(s). These observations are non-isolated; no median, range or ratio is reported. Three qualified fresh pairs in one accepted window are required for timing aggregates.

First public call · observed seconds	Repeated call · observed seconds
21.743564929	6.918791091

Functional source `10b3a9a075fa` · executable `d2cebdd00c2a` · data `4c0877b63b2d`.

Scope. Default prepared 1L–4L caches are attached; no persistent 5L cache is attached before or after either call. On-demand owner preparation is included inside the API timer. Loading, input preparation, serialization, cache snapshots and parity checks are excluded. Reuse may involve several kinds of in-memory state; this is not a selected persistent-cache speedup. Pure IBP, owner-only and terminal-only durations remain unmeasured. These values never enter the cold-process or physical-workload medians. Family identity parity establishes neither catalogue closure nor numerical master availability.

Zero-momentum physical reduction

Completed cache capture · fresh reuse of the same input

Required physical scope: Zero-momentum diagnostic. **Deferred, unmeasured:** Complete quadratic UV fixture. User requested completion with only the zero-momentum physical example; the quadratic example remains deferred until a separate decision. Scope authorization: `b099866588db`. Historical records remain unchanged.

One completed fresh seen-input pair; functional evidence only. Complete retained output, canonical public nonzero-condition file, unknown-master inventory and family map agree. The fresh reuse wrote no new records and preserved the native snapshot identity.

Observed seconds	Capture producer	Fresh seen reuse
Complete public reduction	**144005.8 s**	**5749.0 s**
Supervised collector wall time	145299.8 s	5877.2 s

These are raw single observations, with possible concurrent historical workers. No median, speedup claim or performance colour is assigned. Producer reduction includes capture overhead; pure IBP time is unmeasured. Both processes load the same prepared scalar input; graph preparation is excluded.

Strict records captured	Seen Route hits / misses	Distinct masters / unknowns
2,229,270	**1,750 / 0**	**55 / 0**

Reuse applied 0 rules, with 0 Apply hits, 0 intermediate hits and 0 new records. Captured records are not a deduplicated-key census or the final HEAVY file size.

Producer source `d0e90fbf6537` actual Native archive `ba2a03c60969` ELF `490fd6492677` frozen source `02f29102ca27`. Optimization 3 and fat LTO are bound by the compiler receipt. This original producer cohort is separate from the current application/dependency build. Selected evidence: `physical-capture-717e66516cdcb6f5bd04a30b`. Complete source, executable, input, cache and result bindings remain in `data/results.json`.

Cache production policy. Preserve useful strict roots and completed descendants generated by the physical examples, together with every durable prefix and immutable base. The finite training driver alone does not define the final cumulative cache. Native packs have no 50 MiB publication target or 95 MiB aggregate cap; report actual storage and operational limits.

Seen-input reuse is separate from the independently frozen held-out suite and from same-engine RAM repeats. Protected held-out runs read an immutable snapshot; newly captured work goes to a separate delta. Completed descendants do not prove completion of an unfinished parent, stronger campaign coverage or numerical master availability.

Append accepted evidence with `merge_physical_capture_results.py`. Refresh and validation contract: `maintenance.typ`.

q0 coefficient assembly

Verified engineering observations · 28 October 2026

Coefficient-local retained-result assembly replaces the two whole-expression exact-combination passes in the earlier observation. All 58 master-product coefficients agree exactly with the authenticated original native baseline across 55 distinct masters; all 146678 canonical public guards and the full family map agree. Master values and numerical samples were never evaluated.

Observed seconds	Earlier · dc7364b	New · 2d70bfe Diagnostics OFF
Complete public reduction	6412.7 s	1048.3 s
Prepared input through result serialization	6542.1 s	1194.2 s
Supervised process wall time	6559.5 s	1205.7 s
Retained exact combination*	5815.2 s	231.7 s
Complete native saved call†	565.4 s	770.3 s

The matching prepared-input-to-serialization observations differ by ×5.48 (about 5.5×). **This is a single engineering comparison:** source/cache cohorts, instrumentation and concurrent background work differ. It supplies neither final three-sample medians nor a cache-only speedup claim. Both runs reuse the same prepared scalar input; original graph preparation is excluded.

*Earlier: outer 2941.7 s plus catalogue 2873.4 s. New: coefficient-local catalogue 231.7 s; the outer phase is absent, not a measured zero. †The saved call includes native cache lookup/decode, transport, weighted exact assembly and terminal substitution. Its observed duration increased. Pure IBP and terminal-only durations remain unmeasured.

Diagnostics ON confirms reuse. The separate fresh instrumented process took 1321.3 s through serialization, 1334.5 s supervised wall time, and 1102.8 s for the public call. It returned the identical 784045-byte result and complete guards as OFF, with 1750 Route hits, 0 misses, 0 new rule applications and 0 captured records. OFF/ON controls hit diagnostics; both attach a persistent cache.

New source `2d70bfe439d8` · Native `c2ca815cc906` · kernel `9cd57678d881` · rustc 1.98.1 (48a229cea 2026-09-01) · declared profile release. Original compiled cohorts and verifier-host limits remain in the raw receipts; these checks do not establish final Main-host acceptance.

Selected engineering record: `q0-engineering-5327526dec8cda942773dc57`. Hashed source index: `provenance/q0_engineering_evidence_index.json`. Raw benchmark, execution, exact comparison and all product proofs are appended in `data/results.json`. Earlier selections and all four-loop records remain unchanged.

Cache file engineering observations

HEAVY · one complete cold q0 process. Same executable, source `2d70bfe`, Native `c2ca815` and kernel `9cd5767` as the earlier INDEX run. All 58 coefficients, 55 masters, 146678 canonical public guards and the full result agree. Capture was enabled with delta reads disabled: 1750 Route hits, 0 misses, 0 new rules and 0 captured records.

Observed seconds	Earlier INDEX · OFF	HEAVY single FILE
Data / native cache loading	134.7 / 0.380	135.4 / 931.5
Complete public call	1048.3	777.0
Collector end to end	1194.2	2013.7
Inclusive controller elapsed	1205.7	2132.1
Native saved call / final coefficient combination	770.3 / 231.7	568.6 / 175.2

Public call 25.87% lower; cold collector end to end 68.63% higher. The lower native and CAS observations cannot isolate a cache-format benefit: concurrent work and capture policy differ. No causal format advantage, overall cold speedup, warm result or serialized three-run median is established.

Native FILE loading covers authentication, decode and indexing. The collector made two admitted SHA scans of the 96 GB file: its end-to-end timer includes the first scan and flush/snapshot, excludes close and final SHA; the controller includes both scans and cleanup. Their difference is not pure SHA time. SHA-only, decoder-only and pure IBP times are unmeasured. This legacy CLI used `file_identity=None` with Native framing/binding checks and before/after SHA, distinct from typed R3 acceptance.

LIGHT · one frozen tuning case, two utility pilots. A contains 54 Route records; B adds 2048 Apply records. Both fully match the original INDEX canonical result, guards and five families (68 masters, zero unknowns). Neither used held-out inputs.

Observed utility	A · 54 Route	B · 54 Route + 2048 Apply
Compressed bytes / native records	81907579 / 54	89635796 / 2102
Public / saved call (seconds)	470.7 / 453.6	510.1 / 493.7
Data / cache loading (seconds)	131.3 / 1.42	119.8 / 1.69
New rules / Route hits / Apply hits	52789 / 2 / 0	50381 / 2 / 2048
Closed capture records / directory bytes	117059 / 675020658	114225 / 666760498

B used fewer rules but took longer. The earlier full INDEX observation for this same case took 399.3 s and 4960 new rules (receipt `518d4963a340`). These fresh pilots ran alongside unrelated physical work; they do not establish a performance winner. The later user-directed LIGHT freeze is recorded on the delivery page. All new completed capture nodes remain preserved; neither run read its delta. Public timings include capture overhead.

Both native exports passed full reimport/count/ordered-guard checks within 95000000 compressed bytes. Their separate export helper measured encode+write 9.09 / 15.9 s, reimport 0.564 / 0.641 s and exact visitor comparison 2.28 / 8.02 s (A/B). Exporter identity receipts do not imply that the application supplied a typed Native file identity. Complete source cohorts, closed captures, raw phases and timer scopes remain in `data/results.json`. Source hashes in `provenance/cache_file_engineering_evidence_index.json`.

Zero-momentum delivery checkpoint

Frozen LIGHT cache · exact monitored reuse · 28 October 2026

Scope and timing reference. The required physical example is the complete zero-momentum four-rung fixture; the quadratic example is deferred. The user waived three new uncached repetitions. The completed historical uncached public reduction took 144005.8 s (40.0 h), in its original older cohort. It remains a historical reference, not a matched baseline for a new cache-only ratio. The earlier approximately 5.5× assembly observation concerns Vakint's coefficient-local source change; it is separate from cache-format performance.

LIGHT · accepted as sufficient

Strict native records	Compressed size	Decoded size
2102	**89.64 MB**	**104.66 MB**

Candidate B retains 54 Route and 2048 Apply records. Complete Native reimport verified exact coefficients and the original ordered cache-record guards. The user accepted this existing cache and ended selection/tuning; this does not declare B a performance winner. The earlier A/B observations and all completed capture prefixes remain preserved.

Exact lengths: 89635796 compressed / 104661730 decoded bytes. Compressed SHA-256: `708361954e61e853666d5bb637ad0d58a4336385d4bc9f23133d6ed7e3330021`. Publication and actual download/offline validation remain pending in this checkpoint.

Monitored HEAVY reuse · functional acceptance

Exact master identities / coefficients	Heartbeat freshness bound	Observed peak RAM
55 / 58	**2.077 s**	**8.60 GB**

All 146678 exported guard Atoms agree in order, with mass, couplings, normalization and family identities preserved. 1903 OS/monitor samples observed no faults. The heartbeat covered loading, native work, exact combination, serialization and the final SHA scan; native scientific counters may age during opaque phases. All physical workers have exited. This instrumented check supplies no new performance median.

The peak is an observed OS high-water mark, not a bound on every instantaneous allocation. The 50 GB machine-available-memory guard stayed satisfied; minimum observed available RAM was 379.9 GB. Public guard-vector equality is distinct from the original ordered guards inside each Native cache record.

Final HEAVY export: pending. All 12 precious sources are closed, with 3416555 stored record references before deduplication. Final unique count, compressed/decoded lengths, SHA-256 and export/reimport durations remain unmeasured. The previously validated 96 GB snapshot stays preserved; it is not relabeled as this final union.

Evidence: `provenance/q0_delivery_evidence_index.json`. Exact raw receipts and user decisions are appended in `data/results.json`. No master values were evaluated: the 640 declared five-loop masters remain uncomputed.